

1.3 and 1.4

1.3 Populations, Samples, and Sampling Techniques

1.26. The population in a public university is 7,000 students and you want to sample 200 students. How would you sample these students systematically?

1.27. Describe the difference between a statistic and a parameter

1.28. Why is convenient sampling is considered to be non-statistical sampling method?

1.29. Describe how Kris plans to use the systematic sampling method to sample 100 securities listed on the New York Stock Exchange. Assume that the wall street Journal has 2,531 securities listed in the New York Exchange Composite Transactions" in an alphabetical order

1.34. The Human resources manager of a university e-mailed a survey questionnaire to his 150 employees in the company. The sample included 50 employees randomly selected from each of the category of administration, teaching staff, and non-teaching staff. What sampling method was used by the manager? Explain

Section 1.4 : Data Types and Data Measurement Level

1.42 For each of the following, indicate whether the data are cross sectional or time series

- a) quarterly unemployment rates
- b) unemployment rates by state
- c) monthly sales
- d) employment satisfaction data for a company

1.43 What is the difference between qualitative and quantitative data?

1.44. for each of the following variables, indicate the level of data measurement

- a) product rating (1- excellent, 2 = good, 3 =fair, 4 = poor, 5 = very poor)
- b) home ownership (own, rent, other)
- c) College grade point average
- d) marital status (single, married, divorced, other)

1.45 What is the difference between ordinal and nominal data?

1.47 A company's HR department wants to pick a sample of employees from total of 500 employees in the company and ask how they feel about company policies. The employee' list in alphabetical order is used for the study where every 10th employee will be selected to the sample:

- a) what is the population of interest?
- b) what is the sample?
- c) what is the sampling method being used?

- a. A list of its peer companies

1.49 the following information on 15 different real estate investment trusts has been collected. Identify whether the data collected are cross sectional or time series

- a. income distribution by region in 2016
- b. per share funds from operation for the years 2012-2016
- c. number of residential apartment buildings owned on December 31,2015
- d. overall percentage of leased space for the 137 properties in service as of December 31,2015
- e. annual dividend paid to shareholders in 2010-2015